

Name _____ Period _____

Skill 24.1 Exercise 1

The movies dataframe below is composed of the television shows and movies hosted on the Netflix platform in 2019

	type	title	country	release_year	rating	duration
0	Movie	Norm of the ...	United States	missing	PG	91.071
1	Movie	Jandino: Wha...	United Kingdom	2016	R	94.516
2	TV Show	Transformers...	United States	2013	G	1.127
3	TV Show	Transformers...	United States	2016	TV-14	1.68699999999999
4	Movie	#realityhigh...	United States	2017	TV-14	99.248

For each variable in the *movies* DataFrame, indicate whether it is *numerical* or *categorical*

- Type** - Categorical Values like *Movie* or *TV Show* are labels, not numbers
- Title** - Categorical Text identifiers (names of shows/movies)
- Country** - Categorical Country names are categories
- release_year** - Numerical Represents a year (even though one value is missing, the variable itself is numeric)
- Rating** - Categorical Ratings like *PG*, *R*, *TV-14* are labels, not quantities
- Duration** - Numerical Measured as a number (minutes for movies, seasons for shows)

Skill 24.2 Exercise 1

The line of code below prints the different countries associated with the *country* variable from the *movies* DataFrame above.

Code	
<code>print(movies.country.value_counts())</code>	
Output	
United States	2818
India	972
United Kingdom	419
Japan	245
South Korea	199
...	
Romania, United Kingdom	1
Australia, United Kingdom, United Arab Emirates, Canada	1
Germany, United States, Hong Kong, Singapore	1
Mexico, Brazil	1
United Kingdom, Jordan, Qatar, Iran	1
Name: country, Length: 748, dtype: int64	

For each categorical variable you identified in the previous exercise, indicate whether it is ordinal, nominal, or binary

Categorical Variable	Ordinal, nominal, or binary?
Type	Binary Only two possible values: <i>Movie</i> or <i>TV Show</i>

Name _____ Period _____

Title	Nominal Names/labels with no inherent order
Country	Nominal Country names have no ranking or order (even with 748 unique values)
Rating	Ordinal – Maybe... Ratings imply an order of suitability (e.g., G → PG → PG-13 → R → TV-MA)

Skill 24.3 Exercise 1

Below is revised version of the movies DataFrame,

	show_id	type	title	country	release_year	rating	duration	cast_count	est_budget (USD)
0	s1	Movie	Dick Johnson Is Dead	United States	2020	PG-13	90 min	nan	11959075
1	s2	TV Show	Blood & Water	South Africa	2021	TV-MA	2 Seasons	19.0	9436627
2	s3	TV Show	Ganglands	nan	2021	TV-MA	1 Season	9.0	5719277
3	s4	TV Show	Jailbirds New Orleans	nan	2021	TV-MA	1 Season	nan	7881417
4	s5	TV Show	Kota Factory	India	2021	TV-MA	2 Seasons	8.0	27387486

For each numerical variable, indicate whether it is discrete or continuous

release_year	Discrete Years are counted in whole numbers; values “jump” from one year to the next
cast_count	Discrete Counts of people must be whole numbers (you can’t have 8.5 cast members)
est_budget (USD)	Continuous Represents a measured amount of money and can vary over a wide range, conceptually allowing decimals

Skill 24.4 Exercise 1

Below is a list of data types for the variables in the movies Data Frame

show_id	object
type	object
title	object
country	object
release_year	float64
rating	object
duration	object
cast_count	float64
est_budget (USD)	int64

For each variable below, use the *astype()* method to convert it to data type indicated.

release_year	int64	<code>movies["release_year"] = movies.release_year.astype("int64")</code>
cast_count	int64	<code>movies["cast_count"] = movies.cast_count.astype("int64")</code>

Name _____ Period _____

est_budget (USD)	float64	<code>movies["est_budget (USD)"] = movies['est_budget (USD)'].astype("float64")</code>
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Skill 24.5 Exercise 1

Refer to the movies DataFrame

	show_id	type	title	country	release_year	rating	duration	cast_count	est_budget (USD)
0	s1	Movie	Dick Johnson Is Dead	United States	2020	PG-13	90 min	nan	11959075
1	s2	TV Show	Blood & Water	South Africa	2021	TV-MA	2 Seasons	19.0	9436627
2	s3	TV Show	Ganglands	nan	2021	TV-MA	1 Season	9.0	5719277
3	s4	TV Show	Jailbirds New Orleans	nan	2021	TV-MA	1 Season	nan	7881417
4	s5	TV Show	Kota Factory	India	2021	TV-MA	2 Seasons	8.0	27387486

Below are the categorical variables from a previous exercise. Choose the most appropriate *storage data type* for each variable from the following options: binary, category, or string. Justify each choice.

Categorical Variable	Current data type	Storage data type	Justification
Type	object	category	Two types of categories. They are not True/False values or 0/1
Title	object	string	Each title is unique
Country	object	category	There are lots of repeated countries
Rating	object	category	Rating is ordinal and has repeated values

Use the `astype()` method to convert each variable above.

```
movies["type"] = movies["type"].astype("category")
```

```
movies["title"] = movies["title"].astype("string")
```

```
movies["country"] = movies["country"].astype("category")
```

```
movies["rating"] = movies["rating"].astype("category")
```

Name _____ Period _____

Skill 24.5 Exercise 1

Refer to the movies DataFrame

	show_id	type	title	country	release_year	rating	duration	cast_count	est_budget (USD)
0	s1	Movie	Dick Johnson Is Dead	United States	2020	PG-13	90 min	nan	11959075
1	s2	TV Show	Blood & Water	South Africa	2021	TV-MA	2 Seasons	19.0	9436627
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In a previous example, we converted the rating variable to a category data type. In the space below,

- Create a ratings list with the following values: 'NR', 'G', 'PG', 'PG-13', 'R'
- Use the `cat` method to access the categories
- Use the `set_categories` method to set the order of the categories

```
ratings = ['NR', 'G', 'PG', 'PG-13', 'R']
movies['rating'] = movies['rating'].cat.set_categories(ratings, ordered = True)
```

Skill 24.5 Exercise 1

Refer to the movies DataFrame

	show_id	type	title	country	release_year	rating	duration	cast_count	est_budget (USD)
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In a previous example, we converted the rating variable to a category data type and set the order. Unfortunately, this is the best way to organize this data because it implies a relationship that doesn't necessarily exist.

In the space below One-Hot Encode the ratings variable with `pd.get_dummies()`

```
movies = pd.get_dummies(data = movies, columns = ['rating'])
```

What is the purpose of OHE?

One-hot encoding lets models use categorical data without inventing relationships that don't exist.